

CRF: Form Structure and Why $d_s = 3$

K35 as a Coupling Consistency Condition, Hodge Duality of Space and Gauge Fields

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Companion to CRF MatterSplit and CRF ThetaTower v2.

Abstract. Two results connect the differential form structure of $\mathbb{R}^{d_s}$ to the CRF constant lattice.

(1) K35 as a form consistency condition (algebraically exact). K35 states $\mathcal{G}^2/\varepsilon_0 = n(n+1) = 72$. We prove that $n(n+1)$ equals the total form-complexity-weighted count of all differential forms on $\mathbb{R}^{d_s}$:

$$\sum_{k=0}^{d_s} \binom{d_s}{k} k(n-k) = n(n+1) \iff d_s = 3, n = 2^{d_s}$$

This identity holds *only* for $d_s = 3$. It constitutes a *third independent reason* for three spatial dimensions: the coupling constant $\mathcal{G}$ is consistent with the form structure of $\mathbb{R}^{d_s}$ if and only if $d_s = 3$.

(2) Hodge duality of space and gauge sector. The $n = 8$ powerset splits as 1-forms (space, 3 types) and non-1-forms (matter, 5 types). The 2-forms $\{x, y\}, \{x, z\}, \{y, z\}$ are the Hodge dual of the space 1-forms $\{x\}, \{y\}, \{z\}$. This identifies the 2-form matter sector as the gauge sector — the field-strength 2-forms dual to the coordinate 1-forms. Space and its gauge field are related by Hodge duality.

The Form-Complexity Sum

Definition

On $\mathbb{R}^{d_s}$ with $n = 2^{d_s}$ modes, define the *form-complexity-weighted count*:

$$S(d_s) = \sum_{k=0}^{d_s} \binom{d_s}{k} k(n-k)$$

Here $\binom{d_s}{k}$ is the number of k -forms on $\mathbb{R}^{d_s}$, and $k(n-k)$ is the *form complexity* — the product of the form grade and its codimension.

Evaluation for all d_s

d_s	$n = 2^{d_s}$	$S(d_s)$	$n(n+1)$	Equal?
1	2	1	6	No
2	4	10	20	No
3	8	72	72	Yes
4	16	432	272	No
5	32	2320	1056	No

Why only $d_s = 3$?

Theorem 1 (Form consistency condition). $S(d_s) = n(n+1)$ with $n = 2^{d_s}$ if and only if $d_s = 3$.

Proof. Using $\sum_k \binom{d_s}{k} k = d_s 2^{d_s-1}$ and $\sum_k \binom{d_s}{k} k^2 = d_s(d_s-1)2^{d_s-2} + d_s 2^{d_s-1}$:

$$\begin{aligned}
S(d_s) &= n \sum_k \binom{d_s}{k} k - \sum_k \binom{d_s}{k} k^2 \\
&= n \cdot d_s 2^{d_s-1} - [d_s(d_s-1)2^{d_s-2} + d_s 2^{d_s-1}] \\
&= \frac{n d_s (2n - d_s - 1)}{4}
\end{aligned} \tag{1}$$

The condition $S(d_s) = n(n+1)$ with $n = 2^{d_s}$ becomes:

$$\frac{d_s(2n - d_s - 1)}{4} = n + 1 \quad \Rightarrow \quad n = \frac{d_s^2 + d_s + 4}{2(d_s - 2)}$$

For $d_s = 3$: $n = (9 + 3 + 4)/(2) = 16/2 = 8 = 2^3$. ✓

For $d_s = 4$: $n = (16 + 4 + 4)/(4) = 6 \neq 16 = 2^4$.

For $d_s = 2$: denominator = 0 (no solution).

Therefore $d_s = 3$ is the unique positive integer dimension where $S(d_s) = n(n+1)$ with $n = 2^{d_s}$. $\square$

K35 as a Consistency Condition

K35 (gap 0.021%): $\mathcal{G}^2/\varepsilon_0 = n(n+1) = 72$.

From the theorem: $n(n+1) = S(d_s) = \sum_k \binom{d_s}{k} k(n-k)$.

K35 = Form Consistency, algebraically exact at $d_s = 3$

$$\frac{\mathcal{G}^2}{\varepsilon_0} = n(n+1) = \sum_{k=0}^3 \binom{3}{k} k(8-k) = 0 + 21 + 36 + 15 = 72$$

Breakdown by form grade:

k	$\binom{3}{k}$	$k(8-k)$	Contribution	Sector
0	1	0	0	scalar matter
1	3	7	21	space (1-forms)
2	3	12	36	gauge matter (2-forms)
3	1	15	15	pseudoscalar matter
			72	

The coupling constant $\mathcal{G}^2/\varepsilon_0$ equals the total form-complexity-weighted count of all differential forms on $\mathbb{R}^3$. This is satisfied by no other dimension.

Third reason for $d_s = 3$

CRF has three independent reasons for three spatial dimensions:

1. **SO(3) proof** (CRF SO3 Proof): Axioms 0 and 11 uniquely select $d_s = 3$ as the minimum stable rotation group.
2. **Simulation** (CRF Simulator Report): $d_s \rightarrow 3$ is the attractor of the V20.3 dynamics.
3. **K35 consistency** (this document): $\mathcal{G}^2/\varepsilon_0 = n(n+1)$ holds as a form-complexity identity only for $d_s = 3$.

Three different paths — axioms, dynamics, algebra — all select $d_s = 3$.

Hodge Duality: Space and Gauge

The form split

The CRF powerset split (CRF MatterSplit):

$$n = d_s + n_m \quad \Rightarrow \quad 8 = 3 + 5$$

Space: $k = 1$ forms = $\{x\}, \{y\}, \{z\}$ (3 types)

Matter: $k \neq 1$ forms = $\{\emptyset\}, \{x, y\}, \{x, z\}, \{y, z\}, \{x, y, z\}$ (5 types)

Hodge duality

In $\mathbb{R}^{d_s}$, the Hodge star operator maps k -forms to $(d_s - k)$ -forms:

$$\star : \Omega^k(\mathbb{R}^{d_s}) \rightarrow \Omega^{d_s-k}(\mathbb{R}^{d_s})$$

For $d_s = 3$: 1-forms and 2-forms are Hodge dual ($k = 1 \leftrightarrow k = 2$):

$$\star\{x\} = \{y, z\}, \quad \star\{y\} = \{x, z\}, \quad \star\{z\} = \{x, y\}$$

Space–gauge Hodge duality

Space (1-forms $\{x\}, \{y\}, \{z\}$) and **gauge matter** (2-forms $\{y, z\}, \{x, z\}, \{x, y\}$) are related by the Hodge star.

Space is dual to its own gauge field.

In physics:

- 1-forms = coordinate basis vectors (geometry)
- 2-forms = field-strength tensors, e.g. $F_{\mu\nu}$ (gauge field)
- The Hodge duality $\star F = F$ is the self-duality of electromagnetism

The 3 gauge matter types $\{x, y\}, \{x, z\}, \{y, z\}$ are the 3 components of the magnetic pseudovector $\mathbf{B}$ (a 2-form, not a true vector, in 3D). This is the CRF origin of gauge-field matter.

The full form structure

k	Subsets	Count	Spin	Physical role
0	$\emptyset$	1	0^+	scalar matter
1	$\{x\}, \{y\}, \{z\}$	3	1^+	space axes
2	$\{x, y\}, \{x, z\}, \{y, z\}$	3	1^-	gauge/magnetic
3	$\{x, y, z\}$	1	0^-	pseudoscalar
Space (k=1)		3		geometry
Matter (k≠1)		5		1+3+1 under SO(3)

Form Contributions to K35

The space and matter sectors contribute separately to K35:

$$\begin{aligned}
\text{Space (k=1): } & \binom{3}{1} \cdot 1 \cdot (8-1) = 3 \cdot 7 = 21 \\
\text{Matter (k=0,2,3): } & 0 + 3 \cdot 12 + 1 \cdot 15 = 51 \\
\text{Total: } & 72 = \mathcal{G}^2/\varepsilon_0
\end{aligned}$$

The matter sector contributes $51/72 = 17/24 \approx 70.8\%$ of the coupling. The space sector contributes $21/72 = 7/24 \approx 29.2\%$.

Their ratio: $51/21 = 17/7$ (exact rational).

Status

Result	Status	Precision
$\sum_k \binom{d_s}{k} k(n-k) = n(n+1)$ iff $d_s = 3$	Proved	exact
K35 = form complexity identity	Derived (from above)	exact
Third reason for $d_s = 3$	New	algebraic
Hodge duality: space $\leftrightarrow$ gauge	Derived	exact
2-forms = gauge/magnetic sector	Hypothesis (interpretation)	structural
Matter/space ratio = $17/7$	Derived	exact rational

Summary

K35 is not just a numerical identity. It is a *form consistency condition* that selects $d_s = 3$ from all possible spatial dimensions.

The form-complexity sum $\sum_k \binom{d_s}{k} k(n-k) = n(n+1)$ is satisfied *only* when $d_s = 3$ and $n = 2^{d_s} = 8$. This gives the third independent reason for three spatial dimensions, complementing the SO(3) proof and simulation confirmation.

The Hodge star maps space (1-forms) to the gauge matter sector (2-forms). Space and its gauge field are not separate structures — they are dual faces of the same powerset.

The universe has three dimensions
not just because SO(3) is stable,
not just because the simulator converges to $d_s = 3$,
but because only in three dimensions does the coupling constant
equal the total complexity of all the forms that space can carry.

The geometry and the algebra agree on exactly one answer.